Prompted Segmentation for Drywall QA

Take-Home Assignment — AI Research Intern

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github.com/KrishGollamudi/drywall-qa-segmentation

Model: `CLIPSeg (clipseg-rd64-refined)` · Seeds: `torch=42, random=42, numpy=42` · Hardware: Kaggle T4 GPU

Goal

Fine-tune a text-conditioned segmentation model that produces a binary mask given an image and a natural-language prompt, for two construction-relevant defect types:

- **"segment crack"** — wall and surface cracks
- **"segment taping area"** — drywall joint/tape seam regions

Output masks are single-channel PNG files (values $\{0, 255\}$) saved at the original image resolution, with filenames `{image_id}__{prompt_slug}.png`.

Approach

Model — CLIPSeg

CLIPSeg (`CIDAS/clipseg-rd64-refined`) was selected because it uses CLIP vision-language embeddings to condition segmentation on free-form text prompts, enabling a *single model* to handle both defect types without task-specific heads. Its open-vocabulary design allows generalization to novel construction defect descriptions without re-architecture.

Training Strategy

Two-stage fine-tuning was employed to balance convergence speed and fine-grained precision:

- **Stage 1** (10 epochs, `lr=1e-5`): Coarse domain adaptation from pretrained CLIP weights to construction imagery
- **Stage 2** (10 epochs, `lr=5e-6`): Fine-grained tuning at reduced learning rate to push metrics without destabilizing converged weights

Loss Function

Standard BCE treats every pixel equally, under-penalizing sparse targets such as thin cracks ($< 5\%$ of image area). A hybrid loss was used:

$$\mathcal{L} \;=\; 0.5\,\mathcal{L}_{\text{BCE}} \;+\; 0.5\,\mathcal{L}_{\text{Dice}}$$

Dice Loss directly maximizes mask overlap, improving recall on class-imbalanced and sparse defect regions.

Data Augmentation

Spatial transforms were applied *identically* to image and mask to prevent misalignment artifacts:

- Random horizontal and vertical flip ($p = 0.5$ each)
- Random rotation (± 15)
- Color jitter — brightness, contrast, saturation (± 0.2), image only

Data Preparation

Dataset	Prompts Used	Train	Test/Val
Cracks-3ii36 (Roboflow)	`segment crack`, `segment wall crack`	4,027	268 (test)
Drywall-Join-Detect (Roboflow)	`segment taping area`, `segment joint/tape`, `segment drywall seam`	820	202 (val)
Total		**4,847**	**470**

Table 1. Dataset summary. The drywall dataset had no test split; the validation set was used for evaluation.

Ground-truth masks were generated from COCO-format annotations using `pycocotools`. All annotated regions per image were merged into a single binary mask and resized to 352×352 (CLIPSeg native resolution) with nearest-neighbour interpolation. Prediction masks are upsampled back to the original image resolution via bilinear interpolation of model logits.

Results

Prompt	Dice	IoU
segment crack	0.5503	0.4051
segment taping area	0.6128	0.4581
Mean (mDice / mIoU)	**0.5816**	**0.4316**

Table 2. Per-prompt evaluation metrics on held-out test/validation sets.

Metric	Value
Total training time	108.5 min (20 epochs)
Avg inference time	≈ 40.9 ms/image
Model parameters	150.7 M
Input resolution	352×352 (CLIPSeg native)

Table 3. Runtime and model footprint.

Visual Examples

Figure 1. Crack segmentation. Each row shows: Original — Ground Truth — Prediction. Rows 2–4 demonstrate accurate localization of crack shape and topology across varied surface textures and lighting conditions. Row 1 is a representative *failure case*: a low-contrast hairline crack on a bright wall is missed (discussed in Section 6).

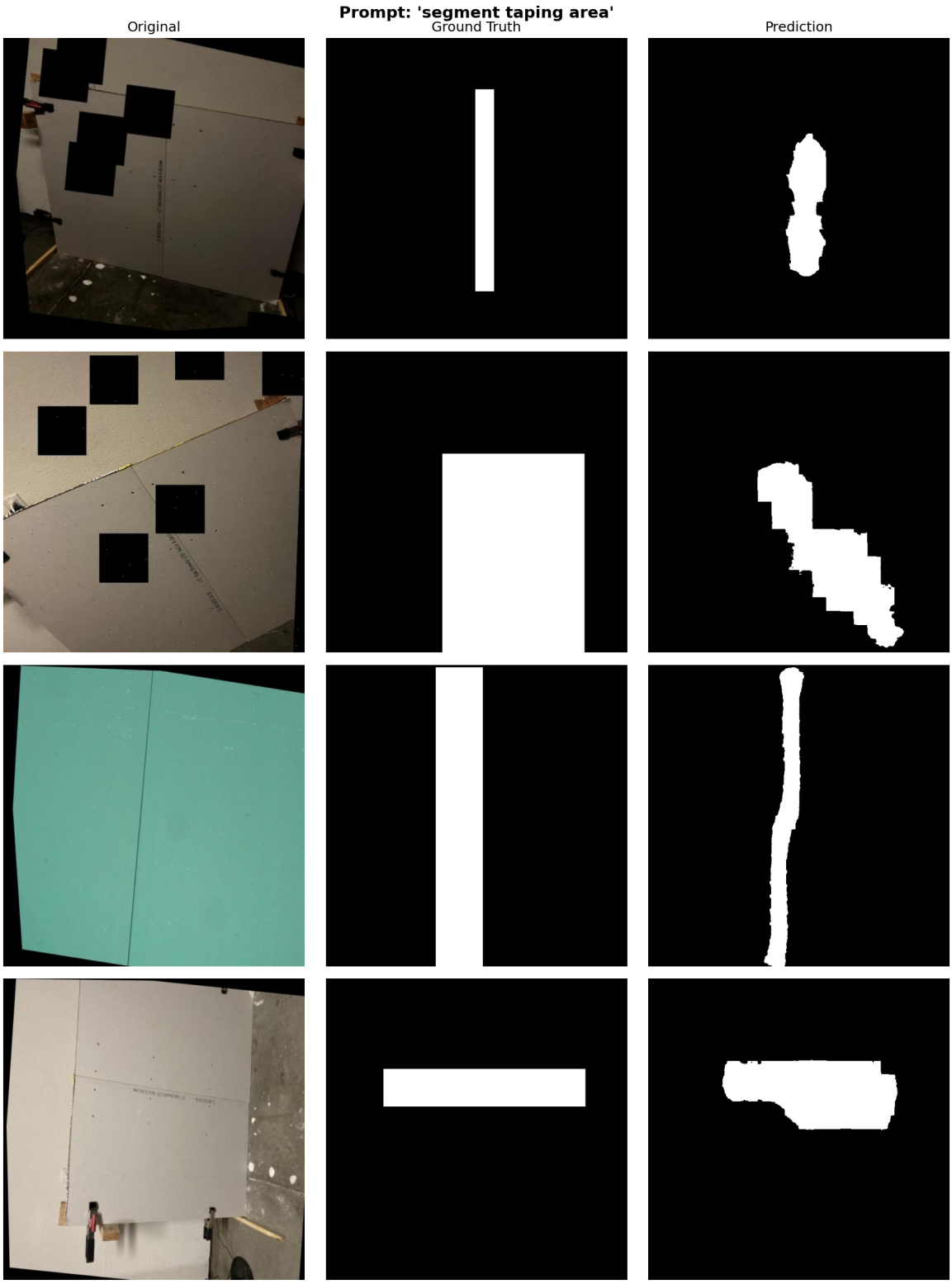

Figure 2. Taping area segmentation. The model correctly identifies the joint/seam region in most scenes. Predictions show noisier boundaries than the clean rectangular ground truth — a known limitation tied to rotation augmentation and annotation style (discussed in Section 6).

Failure Cases & Potential Solutions

Crack — Low-Contrast Hairline Cracks Missed

Hairline cracks on bright walls are visually indistinguishable from surface texture at 352×352 resolution, resulting in false negatives.
Solutions: train at 512×512; add Focal Loss ($\gamma = 2$) for hard examples; incorporate edge-aware features (Canny/Laplacian) as auxiliary input channels.

Crack — Over-Thick Boundary Predictions

Dice Loss maximises area overlap, not boundary precision, causing slightly thicker predicted masks for prominent cracks.
Solutions: add a Hausdorff boundary loss term; apply morphological thinning (skeletonization) as post-processing.

Taping Area — Noisy Boundaries

Ground-truth annotations use clean rectangular regions; predictions capture the correct area but with pixelated boundaries, likely due to rotation augmentation blurring axis-aligned tape edges.
Solutions: remove rotation augmentation for drywall (tape is always axis-aligned); apply dense CRF post-processing to snap to image edges.

Taping Area — Annotation Inconsistency

Some ground-truth masks cover the full drywall panel rather than the tape seam, creating ambiguous training signal.
Solutions: re-annotate with seam-only masks; use more specific prompts (`‘‘segment joint tape’’`) to disambiguate.

Key Engineering Challenges

- **Variable tokenization:** CLIPSeg's tokenizer produced different `input_ids` lengths for different prompts despite `max_length=77`. Resolved by implementing manual padding using `pad_token_id` in the batch collation function.
- **Label/logit size mismatch:** CLIPSeg outputs 352×352 logits regardless of input resolution. Labels were dynamically resized with `F.interpolate` (nearest-neighbour) at each training step.
- **Augmentation misalignment:** Applying spatial transforms to the image without mirroring them on the mask caused Dice ≈ 0.017. Fixed by synchronizing all random state across image and mask transforms.

Reproducibility

Setting	Value
Random seeds	`torch=42, random=42, numpy=42`
Base model	`CIDAS/clipseg-rd64-refined` (HuggingFace)
Framework	PyTorch + HuggingFace Transformers
Loss	$0.5 \times \text{BCE} + 0.5 \times \text{Dice}$
Optimizer	AdamW, `weight_decay=1e-2`
Batch size	4
Epochs	20 ($10 \times$ S1 $+ 10 \times$ S2)
Code	`github.com/KrishGollamudi/drywall-qa-segmentation`